

Alexander Gabriel Aranes

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SUMMARY

A full-stack freelance software developer and AI researcher with experience with different tech stacks from creating projects in various platforms. His skills involves web development tools like MERN, FastAPI, and NextJS. He conducted a research highlighting his knowledge with data science tools like pandas, scikit-learn, and PyTorch. He is also well versed in game and app development using ElectronJS, Flutter and Godot.

SKILLS

Languages: Python, TypeScript, C, Java, R, C#, Rust, Dart

Software Development : NextJS, MERN, Streamlit, Godot, Flutter, Dreamflow, Aseprite

Data Science, AI & ML : Numpy, Pandas, Scikit-learn, PyTorch, Google Colab

Systems and Infrastructure: AWS, Google Cloud, Git, Linux,

EXPERIENCE

Freelance Software Developer August 2023 – present

Project Commissions

- Created a full game in Java using JavaFX from Animation threading to game logic.
- Has experience programming code for an Arduino UNO sketch for a capstone project that uses an automatic watering system. The project was monitored inside Philippines Rice Research Institute (PhilRice).
- Created a full-stack E-commerce web-application using ASP.NET framework.

Indie Game Developer January 2024 – present

Game Jams

- Created hyper-casual games using Godot Game engine that were submitted to game jams at Itch.io.
- Participated in *Jame Gam #50*, *Game Jam Los Baños 2024*, *Game Jam Los Baños 2025*, *GameJamPlus 24/25*.

Student Researcher April 2025 – present

Machine Learning and Artificial Intelligence Application Lab (MALAIA Lab)

- Conducted a research regarding Filipino's first salary after graduating
- Trained different classical machine learning models using LIYAB dataset
- Conducted an exploratory data analysis that helped on feature engineering for a more robust set of data
- Preliminary qualifier at 9th International Conference on Machine Learning and Machine Intelligence (MLMI 2026)

Backend Lead Developer February 2026 – May 2026

The Haven Company: Undergraduate Course Requirement

- Applied a comprehensive Backend SOP to the project's pull requests. Reviewed NextJS backend code utilizing Zod Schematics and OpenAPI documentation
- Handled a full-stack team integrating a feature every sprint of the project timeline
- Created a curated backend learning material for up-skilling

LEADERSHIP

Capstone Research Head March 2022 - June 2023

Senior High School Capstone Research

- Delegated programming and research tasks to co-researchers. Became the lead programmer for an automated waste sorter machine powered with Arduino Nano and Orange Pi One mini-PC.

Affiliate's Internal Event Head Sep 2025 - Nov 2025

Alliance of Computer Science Students - UPLB (ACSS-UPLB) Internal Event

- Managed an ad-hoc team divided into divisions to deliver a tutorial regarding Automata and Language theory.

Project Manager September 2025 - November 2025

PJDSC 2025

- Lead the contextual inquiry and full-stack development of SAGIP, a two-modal PWA/web-application with early warning features that communicates the LGU to its community before a disaster.

Outreach Program Overseer November 2025

Paghamon sa ISIPnayan

- Oversaw the implementation phase of NSTP - LTS project.
- Lead program planning and content formulation of teaching materials.

Hackathon Assistant Head *November 2025 - March 2026*

The Innovation Lab 2026 - ACSS Week 2026

- Handled secretarial tasks and provided consultations for the event's division heads for key major decisions.
- Performed external relation tasks by coordinating with the event speakers and judges.

ACHIEVEMENTS

Top BSCS Recognition *March 2024 & April 2025*

- Recognized by the Institute of Computer Science, CAS, UPLB as Top 1 Freshman and Top 3 Sophomore during the Academic Years 2023-2024 and 2024-2025 respectively.

2nd Runner Up, DOST Hack4AProgress 2025 *July 2025*

- A premier regional 48-hour hackathon designed to create innovative software solutions for local government challenges,

2nd Runner Up, Philippine Junior Data Science Challenge 2025 *November 2025*

- A flagship national competition for students organized by the UP Data Science Society to create data-driven solutions for societal issues

Honor Society of the Philippines Invitation *March 2026*

- Recognized and exclusively invited by the Honor Society of the Philippines (HSP), the first Philippine-based academic honor society, as one of the top 7.5% juniors from University of the Philippines.

CERTIFICATIONS

DataCamp AI Fundamentals Certification *January 2025*

DOST-SEI Scholar's Leadership Camp - Patriot Scholar Certificate *May 2026*

PROJECTS

Budget Utilization and Local Spending Analyst (BULSA) *July 2025*

- A web application built in NextJS and FastAPI utilizing vector embeddings through a RAG pipeline for processing LGU's financial documents. Submitted to DOST Hack4AProgress 2025

BILL AI *November 2025*

- An agentic bill tracker that utilizes RAG-pipeline powered with OpenAI api to streamline document processing. Submitted to CodeKada: The Innovation Hackathon 2025.

Online Secret Santa Platform *December 2025*

- Serverless platform for secret santa deployed using vercel.

Academic Companion (ACOM) *January 2026*

- An academic status tracker built in Flutter using Dreamflow. Submitted to Dreamflow's Buildathon 2025.

EDUCATION

University of the Philippines Los Baños *August 2023 – August 2027*

Bachelor of Science - BS, Computer Science, GWA 1.24, Dept. of Science and Technology Undergraduate Scholar

Laguna Science Integrated High School *June 2017 – July 2023*

With High Honors, High School Diploma